

THE SUPER WEIGHT IN LARGE LANGUAGE MODELS

<https://arxiv.org/pdf/2411.07191>

Abstraction:

1.

In this work, we present an even more surprising finding: Pruning as few as a single parameter can destroy an LLM’s ability to generate text – increasing perplexity by 3 orders of magnitude and reducing zero-shot accuracy to guessing.

2. We propose a datafree method for identifying such parameters, termed super weights, using a single forward pass through the model.

3. We additionally find that these super weights induce correspondingly rare and large activation outliers, termed super activations. When preserved with high precision, super activations can improve simple round-to-nearest quantization to become competitive with state-of-the-art methods.

4. For weight quantization, we similarly find that by preserving the super weight while clipping other weight outliers, round-to-nearest quantization can scale to much larger block sizes than previously considered.

INTRODUCTION



Figure 1: Super Weight Phenomenon. We discover that pruning a single, special scalar, which we call the *super weight*, can completely destroy a Large Language Model’s ability to generate text. On the left, the original Llama-7B, which contains a super weight, produces a reasonable completion. On the right, after pruning the super weight, Llama-7B generates complete gibberish. As we show below, this qualitative observation has quantitative impact too: zero-shot accuracy drops to guessing and perplexity increases by orders of magnitude.

Our contributions are summarized as follows.

1. **Super Weights:** We discover a tiny subset of outliers in LLMs, at most six scalars, that are disproportionately important; pruning these super weights destroys model quality.
2. **Identifying Super Weights:** We present a data-free way to identify super weights using only a single forward pass and provide an index of super weights for existing, open LLMs.
3. **Super Activations:** We analyze how super weights influence inference and relate them to the activation outliers observed in prior work.
4. **Compression:** By preserving super outliers, we show that round-to-nearest quantization increases effectiveness noticeably; preserving super outliers improves compression quality.

SUPER WEIGHTS

1. a small percentage of the largest magnitude outliers are essential to model quality. This percentage can be as small as 0.01%, but for these billion-parameter models, 0.01% can still include hundreds of thousands of weights.
2. Our investigation reveals a surprising fact about even this group of weight outliers: There exists a single, scalar weight that, despite not being the largest, holds more importance than thousands of other outlier weights combined. We call this single scalar weight a super weight.

In our analysis, we find that the super weight is necessary for quality, having an outsized influence on quality if removed.

Model	No.	Type	Weight	Coordinates
Llama 7B	2	mlp	down_proj	[3968, 7003]
Llama 13B	2	mlp	down_proj	[2231, 2278]
	2	mlp	down_proj	[2231, 6939]
Llama 30B	3	mlp	down_proj	[5633, 12817]
	3	mlp	down_proj	[5633, 17439]
	10	mlp	down_proj	[5633, 14386]
Llama2 7B	1	mlp	down_proj	[2533, 7890]
Llama2 13B	3	mlp	down_proj	[4743, 7678]
Mistral-7B v0.1	1	mlp	down_proj	[2070, 7310]

Model	No.	Type	Weight	Coordinates
OLMo-1B 0724-hf	1	mlp	down_proj	[1764, 1710]
	1	mlp	down_proj	[1764, 8041]
OLMo-7B 0724-hf	1	mlp	down_proj	[269, 7467]
	2	mlp	down_proj	[269, 8275]
	7	mlp	down_proj	[269, 453]
	24	mlp	down_proj	[269, 2300]
Phi-3 mini-4k-instruct	2	mlp	down_proj	[525, 808]
	2	mlp	down_proj	[1693, 808]
	2	mlp	down_proj	[1113, 808]
	4	mlp	down_proj	[525, 2723]
	4	mlp	down_proj	[1113, 2723]
	4	mlp	down_proj	[1693, 2723]

Table 2: Super Weight Directory. The above layer numbers, layer types, and weight types can be directly applied to Huggingface models. For example, for Llama-7B on Huggingface, access the super weight using `layers[2].mlp.down_proj.weight[3968, 7003]`.

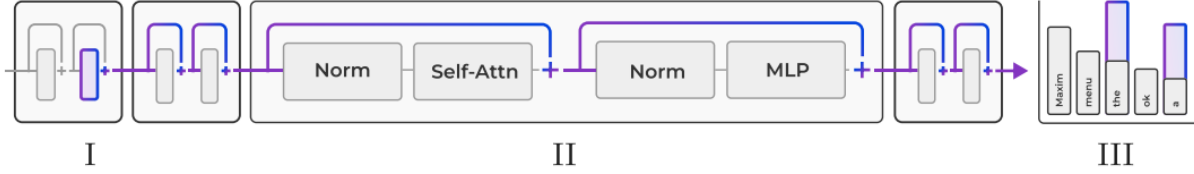


Figure 2: How Super Weights behave. *I:* Super weights are often found in an early layer’s down projection, indicated with a blue-purple box. The super weight immediately creates an incredibly large-magnitude super activation. *II:* Super activations are propagated through skip connections, indicated with blue-purple lines. *III:* This has a net effect of suppressing stopword likelihoods in the final logits. Removing the super weight causes stopword likelihood to increase, indicated with the blue-purple stacked bars. See Appendix A.3.

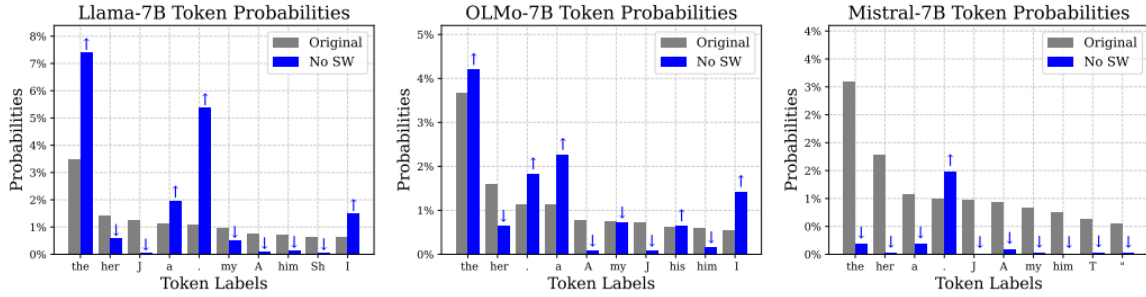


Figure 5: Super weights suppress stopwords. Above, we consistently observe that removing super weights results in 2-5× larger stopword probabilities, across a variety of LLMs. At the same time, we observe non-stopwords decrease sharply in probability, reducing by 2-3× to as little as 0.1% probability. Overall, this results in stopwords dominating the highest likelihood tokens.

SUPER-OUTLIER AWARE QUANTIZATION

quantization critical. Quantization generally maps continuous values to a finite set of values; we consider one of the simplest forms – namely, asymmetric round-to-nearest quantization:

$$Q(\mathbf{X}) = \text{Round} \left(\frac{\mathbf{X} - \text{MIN}(\mathbf{X})}{\Delta} \right), Q^{-1}(\hat{\mathbf{X}}) = \Delta \cdot \hat{\mathbf{X}} + \text{MIN}(\mathbf{X})$$

where $\Delta = \frac{\text{MAX}(\mathbf{X}) - \text{MIN}(\mathbf{X})}{2^{N-1} - 1}$ is the quantization step and N is the number of bits. Note that the maximum value is used to calculate Δ , so super outliers in X drastically increase the step size. With larger step sizes, inliers are rounded to more distant values on average, increasing the quantization error. With increasingly super outliers, inliers are rounded to fewer discrete values, and more

PPL ($\downarrow$)	Llama-7B		Llama-13B		Llama-30B	
	Wiki-2	C4	Wiki-2	C4	Wiki-2	C4
FP16	5.68	7.08	5.09	6.61	4.10	5.98
Naive W8A8	5.83 (0%)	7.23 (0%)	5.20 (0%)	6.71 (0%)	4.32 (0%)	6.14 (0%)
SmoothQuant	5.71 (100%)	7.12 (100%)	5.13 (100%)	6.64 (100%)	4.20 (100%)	6.06 (100%)
Ours	5.74 (75%)	7.14 (82%)	5.15 (71%)	6.66 (71%)	4.22 (83%)	6.08 (75%)

Table 3: Round-to-nearest with super-activation handling is competitive. W8A8 is the baseline 8-bit weight and activation quantization, and the small italicized, parenthesized percentages denote what percentage of SmoothQuant’s quality improvement is retained. We observe that a naive round-to-nearest, while handling a single scalar super activation per tensor, is competitive with SmoothQuant. Note that SmoothQuant uses calibration data to compute scales, whereas our method does not require data.

4.1 ACTIVATION QUANTIZATION

We conduct experiments using round-to-nearest quantization, with a small modification – replace the super activation with the median value (REPLACE), quantize (Q) and dequantize (Q^{-1}) activations, then restore the super activation in FP16 (RESTORE). This can be expressed as the following,

$$\hat{A} = \text{RESTORE}(Q^{-1}(Q(\text{REPLACE}(A)))) \quad (1)$$

Since the super activation is a single scalar, the bitrate and kernel complexity are not significantly impacted.

4.2 WEIGHT QUANTIZATION

$$\hat{W} = \text{RESTORE}(Q^{-1}(Q(\text{CLIP}_z(W)))) \quad (2)$$

EXPERIMENTS

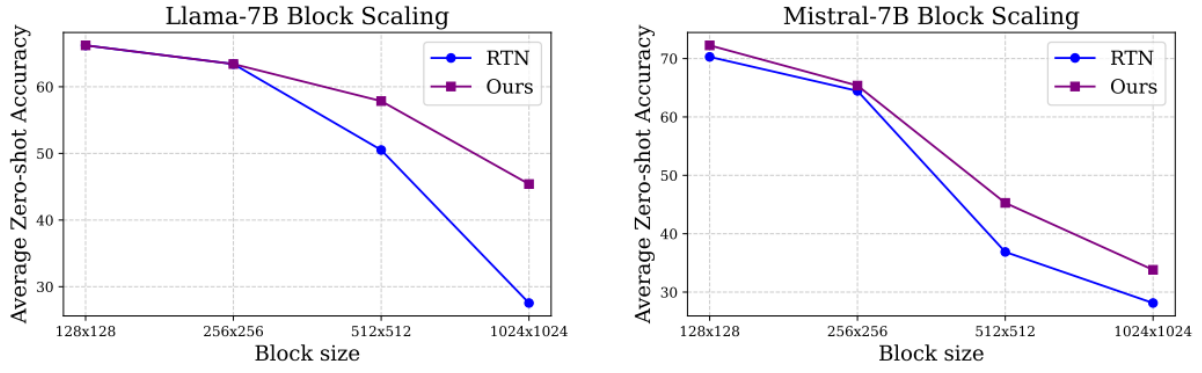


Figure 7: Restoring super weight improves block scaling. Smaller block sizes are often used to handle outliers implicitly. We note that block sizes can scale slightly more gracefully by just handling the single scalar-valued super weight.